

Exercises

- 1.⁻ Show that the four colour theorem does indeed solve the map colouring problem stated in the first sentence of the chapter. Conversely, does the 4-colourability of every map imply the four colour theorem?
 - 2.⁻ Show that, for the map colouring problem above, it suffices to consider maps such that no point lies on the boundary of more than three countries. How does this affect the proof of the four colour theorem?
 3. Try to turn the proof of the five colour theorem into one of the four colour theorem, as follows. Defining v and H as before, assume inductively that H has a 4-colouring; then proceed as before. Where does the proof fail?
 4. Calculate the chromatic number of a graph in terms of the chromatic numbers of its blocks.
 - 5.⁻ Let G be a graph, and let $k \in \mathbb{N}$.
 - (i) Show that G has chromatic number at most k if and only if there exists a homomorphism from G to K^k .
 - (ii) Show that G is bipartite if and only if there exists a homomorphism from G to K^2 or to an even cycle.
 - (iii) Are there homomorphisms from C^{17} to C^7 , from C^7 to C^{17} , from C^{16} to C^7 , and from C^{17} to C^6 ?
 6. Show that graphs of large girth and without a given minor are ‘nearly bipartite’ in the following sense. Let H be a fixed graph and C a fixed odd cycle. Use Theorem 7.2.6 to show that if G is a graph of sufficiently large girth (depending only on H and C) that does not contain H as a minor, then there is a homomorphism from G to C .
 7. For every $n > 1$, find a bipartite graph on $2n$ vertices, ordered in such a way that the greedy algorithm uses n rather than 2 colours.
 8. Consider the following approach to vertex colouring. First, find a maximal independent set of vertices and colour these with colour 1; then find a maximal independent set of vertices in the remaining graph and colour those 2, and so on. Compare this algorithm with the greedy algorithm: which is better?
 9. Show that the bound of Proposition 5.2.2 is always at least as sharp as that of Proposition 5.2.1.
- A k -chromatic graph G is called *critically k -chromatic*, or just *critical*, if $\chi(G - v) < k$ for every $v \in V(G)$.
10. Determine the critical 3-chromatic graphs.
 - 11.⁺ Show that every critical k -chromatic graph is $(k - 1)$ -edge-connected.
 12. Formalize and prove the following statement: assuming large average degree drives the colouring number up but not the chromatic number.

13. Write $\text{col}'(G)$ for the least number of colours used by the greedy algorithm for a suitable vertex ordering of a graph G . Does every G satisfy $\text{col}'(G) = \text{col}(G)$ or $\text{col}'(G) = \chi(G)$?
14. Find a function f such that every graph of arboricity at least $f(k)$ has colouring number at least k , and a function g such that every graph of colouring number at least $g(k)$ has arboricity at least k , for all $k \in \mathbb{N}$.
15. Given $k \in \mathbb{N}$, find a constant $c_k > 0$ such that every large enough graph G with $\alpha(G) \leq k$ contains a cycle of length at least $c_k |G|$.
16. Find a graph G for which Brooks's theorem yields a significantly weaker bound on $\chi(G)$ than Proposition 5.2.2.
17. Show that, in order to prove Brooks's theorem for a graph $G = (V, E)$, we may assume that $\kappa(G) \geq 2$ and $\delta(G) \geq 3$. Then prove the theorem under these assumptions, showing first the following two lemmas.
 - (i) Let $v_1, \dots, v_n$ be an enumeration of V . If every v_i ($i < n$) has a neighbour v_j with $j > i$, and if $v_1 v_n, v_2 v_n \in E$ but $v_1 v_2 \notin E$, then the greedy algorithm uses at most $\Delta(G)$ colours.
 - (ii) If G is not complete, it has a vertex v_n with non-adjacent neighbours v_1, v_2 that do not separate G .
18. Show that the following statements are equivalent for a graph G :
 - (i) $\chi(G) \leq k$;
 - (ii) G has an orientation without directed paths of length k ;
 - (iii) G has an acyclic such orientation (one without directed cycles).
19. Given a graph G and $k \in \mathbb{N}$, let $P_G(k)$ denote the number of vertex colourings $V(G) \rightarrow \{1, \dots, k\}$. Show that P_G is a polynomial in k of degree $n := |G|$, in which the coefficient of k^n is 1 and the coefficient of k^{n-1} is $-||G||$. (P_G is called the *chromatic polynomial* of G .)
(Hint. Apply induction on $||G||$.)
20. Determine the class of all graphs G for which $P_G(k) = k(k-1)^{n-1}$. (As in the previous exercise, let $n := |G|$, and let P_G denote the chromatic polynomial of G .)
21. Show that for every $k \in \mathbb{N}$ there is a unique $\subseteq$ -minimal 'Kuratowski class' $\mathcal{X}_k$ of k -chromatic graphs such that every k -chromatic graph has a subgraph in $\mathcal{X}_k$, but that for $k \geq 3$ this class $\mathcal{X}_k$ is never finite.
22. For every $k \in \mathbb{N}$, construct a triangle-free k -chromatic graph. Can you even make them C^4 -free?
23. Consider an infinite matrix whose rows and columns are both indexed by $\mathbb{N} \setminus \{0\}$. For all $i, j \in \mathbb{N} \setminus \{0\}$ join the element v_{ij} in row i and column j to every element of column $i+j$. Prove that the resulting graph contains no triangle. What is its chromatic number?

24. In the definition of ' k -constructible', replace axioms (ii) and (iii) by
- (ii)' Every supergraph of a k -constructible graph is k -constructible.
 - (iii)' If x, y_1, y_2 are distinct vertices of a graph G and $y_1y_2 \in E(G)$, and if both $G + xy_1$ and $G + xy_2$ are k -constructible, then G is k -constructible.

Show that a graph is k -constructible with respect to this new definition if and only if its chromatic number is at least k .

25. ⁻ An $n \times n$ -matrix with entries from $\{1, \dots, n\}$ is called a *Latin square* if every element of $\{1, \dots, n\}$ appears exactly once in each column and exactly once in each row. Recast the problem of constructing Latin squares as a colouring problem.
26. Without using Proposition 5.3.1, show that $\chi'(G) = k$ for every k -regular bipartite graph G .
27. Prove Proposition 5.3.1 from the statement of the previous exercise.
28. ⁺ A forest is called *linear* if all its components are paths. Show that every cubic graph has an edge-decomposition into two linear forests.
29. ⁻ Without using Theorem 5.4.2, show that every plane graph is 6-list-colourable.
30. For every integer k , find a 2-chromatic graph whose choice number is at least k .
31. ⁻ Find a general upper bound for $\text{ch}'(G)$ in terms of $\chi'(G)$.
32. Compare the choice number of a graph with its colouring number: which is greater? Can you prove the analogue of Theorem 5.4.1 for the colouring number (directly, without using any major theorem)?
33. ⁺ Prove that the choice number of K_2^r is r .
34. The *total chromatic number* $\chi''(G)$ of a graph $G = (V, E)$ is the least number of colours needed to colour the vertices and edges of G simultaneously so that any adjacent or incident elements of $V \cup E$ are coloured differently. The *total colouring conjecture* says that $\chi''(G) \leq \Delta(G) + 2$. Bound the total chromatic number from above in terms of the list-chromatic index, and use this bound to deduce a weakening of the total colouring conjecture from the list colouring conjecture.
35. ⁻ Does every oriented graph have a kernel? If not, does every graph have an orientation in which every induced subgraph has a kernel? If not, does every graph have an orientation that has a kernel?
36. ⁺ Prove that every directed graph without odd directed cycles has a kernel.
37. Show that every bipartite planar graph is 3-list-colourable.
(Hint. Apply the previous exercise and Lemma 5.4.3.)
38. ⁻ Show that perfection is closed neither under edge deletion nor under edge contraction.

- 39.⁻ Deduce Theorem 5.5.6 from the strong perfect graph theorem.
40. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two sets of imperfect graphs, each minimal with the property that a graph is perfect if and only if it has no induced subgraph in $\mathcal{H}_i$ ($i = 1, 2$). Do $\mathcal{H}_1$ and $\mathcal{H}_2$ contain the same graphs, up to isomorphism?
41. Use König's Theorem 2.1.1 to show that the complement of any bipartite graph is perfect.
42. Using the results of this chapter, find a one-line proof of the following theorem of König, the dual of Theorem 2.1.1: in any bipartite graph without isolated vertices, the minimum number of edges meeting all vertices equals the maximum number of independent vertices.
43. A graph is called a *comparability graph* if there exists a partial ordering of its vertex set such that two vertices are adjacent if and only if they are comparable. Show that every comparability graph is perfect.
44. A graph G is called an *interval graph* if there exists a set $\{I_v \mid v \in V(G)\}$ of real intervals such that $I_u \cap I_v \neq \emptyset$ if and only if $uv \in E(G)$.
- Show that every interval graph is chordal.
 - Show that the complement of any interval graph is a comparability graph.
- (Conversely, a chordal graph is an interval graph if its complement is a comparability graph; this is a theorem of Gilmore and Hoffman (1964).)
45. Show that $\chi(H) \in \{\omega(H), \omega(H) + 1\}$ for every line graph H .
- 46.⁺ Characterize the graphs whose line graphs are perfect.
47. Let G' be obtained from a graph G by replicating a vertex. If G satisfies $\chi = \omega$, does G' too?
48. Show that a graph G is perfect if and only if every non-empty induced subgraph H of G contains an independent set $A \subseteq V(H)$ such that $\omega(H - A) < \omega(H)$.
- 49.⁺ Consider the graphs G for which every induced subgraph H has the property that every maximum-size complete subgraph of H meets every maximum-size independent vertex set in H .
- Show that these graphs G are perfect.
 - Show that these graphs G are precisely the graphs not containing an induced copy of P^3 .
- 50.⁺ Show that in every perfect graph G one can find a set $\mathcal{A}$ of independent vertex sets and a set $\mathcal{K}$ of vertex sets of complete subgraphs such that $\bigcup \mathcal{A} = V(G) = \bigcup \mathcal{K}$ and every set in $\mathcal{A}$ meets every set in $\mathcal{K}$.
(Hint. Lemma 5.5.5.)
- 51.⁺ Let G be a perfect graph. As in the proof of Theorem 5.5.4, replace every vertex v of G with a perfect graph G_v (not necessarily complete), adding all $G_u - G_v$ edges whenever uv is an edge of G . Show that the resulting graph G' is again perfect.

52. Show that $\chi(G) \geq \max \{ \omega(G), \lceil |G|/\alpha(G) \rceil \}$ for all graphs G .
- (i)⁻ Find an example of G where this inequality is strict.
 - (ii)⁺ Prove that all graphs G obtained from cycles by repeated vertex replication satisfy it with equality.
- 53.⁻ Discuss why χ -boundedness is defined only for graph properties that are closed under induced subgraphs. Support your arguments by the construction of a class that satisfies the definition of χ -boundedness but is *not* closed under induced subgraphs, so that the graphs in this class are very similar to the triangle-free graphs.
- 54.⁻ Reduce the Gyárfás-Sumner conjecture to trees: prove that if not containing any fixed tree as an induced subgraph is a χ -bounded property then so is that of not containing any fixed forest.

Notes

The authoritative reference work on all questions of graph colouring is T.R. Jensen & B. Toft, *Graph Coloring Problems*, Wiley 1995. Starting with a brief survey of the most important results and areas of research in the field, this monograph gives a detailed account of over 200 open colouring problems, complete with extensive background surveys and references. Most of the remarks below are discussed comprehensively in this book, and all the references for this chapter can be found there. A book specifically on edge colouring is L.M. Favrholt, D. Scheide, M. Stiebitz & B. Toft, *Graph Edge Coloring: Vizing's Theorem and Goldberg's Conjecture*, Wiley 2012.

The *four colour problem*, whether every map can be coloured with four colours so that adjacent countries are shown in different colours, was raised by a certain Francis Guthrie in 1852. He put the question to his brother Frederick, who was then a mathematics undergraduate in Cambridge. The problem was first brought to the attention of a wider public when Cayley presented it to the London Mathematical Society in 1878. A year later, Kempe published an incorrect proof, which was in 1890 modified by Heawood into our first proof of the five colour theorem. In 1880, Tait announced 'further proofs' of the four colour conjecture, which never materialized; see the notes for Chapter 10. Our second proof of the five colour theorem dates back at least to 1972, when Woodall used it in his first graph theory course.

The first widely accepted proof of the four colour theorem was published by Appel and Haken in 1977. The proof builds on ideas that can be traced back as far as Kempe's paper, and were developed largely by Birkhoff and Heesch. Very roughly, the proof sets out first to show that every plane triangulation must contain at least one of 1482 certain 'unavoidable configurations'. In a second step, a computer is used to show that each of those configurations is 'reducible', i.e., that any plane triangulation containing such a configuration can be 4-coloured by piecing together 4-colourings of smaller plane triangulations. Taken together, these two steps amount to an inductive proof that all plane triangulations, and hence all planar graphs, can be 4-coloured.

Appel & Haken's proof has not been immune to criticism, not only because of their use of a computer. The authors responded with a 741 page long